

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
6	(a)		1 clockwise arrow (1) 1, 2, 3, 4 in correct positions 1(1) 42 3		2		2		
	(b)		True because clockwise OR process 1 has greatest work than 3 (1) No work done for 2 & 4 (1) Apply ecf if needed			2	2		
	(c)		Attempt made at any area under the graph or enclosed or counting squares (1) 9.25ish large squares or 9×25 small squares or correct trapezium / enclosed area method (1) Answer = $4\,600 \pm 400$ [J] (1)			3	3	3	
	(d)		Heat flows from gas to surroundings (through piston) OR conduction through metal (1) Accept molecules collide with container [Thermal] equilibrium or same temperature as surroundings (1)		1	1	2		
	(e)		Any 4 × (1) from: • Volume increases [for expansion part] • Work done by gas [for expansion but not compression] • Internal energy decrease [for expansion but not compression] • Increased PE linked to lower temperature or expansion • KE or internal energy decrease linked to cooling or lower temperature or expansion • [Process] too quick for heat transfer			4	4		